

Sultan Muhammad Khan

AI Engineer | Game Developer

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aka-shiro10.github.io/Sultan-portfolio/ | github.com/aka-shiro10

Summary

I specialize in Artificial Intelligence with hands-on experience in machine learning, game AI, and applied intelligent systems. Most of my work focuses on Python and Unity, where I build and experiment with intelligent agents, simulations, and practical AI-driven systems. I prefer learning-by-building: rapid prototyping, testing, and refining working systems.

Education

BS in Artificial Intelligence

Oct 2022 – Present

Pak-Austria Fachhochschule Institute of Applied Sciences and Technology

Intermediate (ICS)

2020 – 2022

Jinnah Jame School & College, Haripur

Matriculation

2018 – 2020

Jinnah Jame School & College, Haripur

Industrial Training & Internships

Data / ML Intern — Sybrid

May 2025 – Jul 2025

Wah Cantt / Remote

- Curated datasets for ETL pipelines, stood up a vector database, and prototyped an internal RAG-based chatbot (deployment-ready proof-of-concept).

Software Intern — KeyMech

Jul 2024 – Sep 2024

Wah Cantt

- Delivered cross-platform mobile features, integrated REST APIs, and prepared production builds for field maintenance teams.

Audit Intern — Seasons Foods

Aug 2023 – Sep 2023

Lahore

- Performed inventory audits and reconciliations and suggested process improvements that reduced count variances.

Final Year Project

AI-Powered Smart System

Sep 2024 – Present

An integrated, real-time system combining machine learning, computer vision, and NLP to address practical problems. Responsibilities include dataset preparation, model training and optimization, UI prototyping, and system integration. (Team size: 2–3)

Technical Skills

AI / ML: Machine Learning, Deep Learning, Computer Vision, NLP, Reinforcement Learning

Game & Simulation: Unity (C#), ML-Agents, Agent Behavior Systems, Networking

Languages & Tools: Python, C#, C/C++, JavaScript, Git, SQL, Docker, Azure

Frameworks & Platforms: TensorFlow, PyTorch (basic), React, Tailwind, WordPress, WooCommerce

Hardware / Embedded: Arduino, QMK (macropad)

Selected Projects

- **Evolution Simulator** — Unity simulation using artificial neural networks and genetic algorithms to evolve agent behaviors under dynamic environmental food conditions. (Unity, ANN, Genetic Algorithms)
- **Muffin Time** — Networked Unity platform game featuring handcrafted art and real-time multiplayer mechanics. (Unity, Networking, Photoshop)
- **Car Racing AI Trainer** — Unity + ML-Agents project training RL agents to learn driving and racing strategies on procedural tracks. (Unity, ML-Agents)
- **Network Anomaly Detection** — Python-based ML system that detects anomalous network traffic and presents results in a lightweight GUI. (Python, ML, GUI)
- **Intrusion Detection System** — Packet analysis and ML-based intrusion detection prototype for detecting suspicious network activity. (Python, ML, Security)
- **Server Parallel Log Analyzer** — High-performance Python tool for concurrent processing of massive server logs with real-time statistics and parallel computing. (Python, Parallel Computing)
- **Desktop AI Pet** — Interactive desktop companion using computer vision and NLP to respond to users and perform playful tasks. (Python, CV, NLP)
- **Text-to-Speech** — Python TTS module that converts text into natural-sounding speech for accessibility and prototypes. (Python)
- **Chess vs AI** — Python chess game with a minimax and heuristic-based AI opponent for single-player practice. (Python)
- **Snake Game** — Classic Snake implemented in Python/Pygame with scoring and simple level mechanics. (Python, Pygame)
- **Flabby Bird** — Flappy-Bird-style arcade game in Python with Pygame and SQLite-based leaderboard persistence. (Python, Pygame, SQLite)
- **Media Player** — Python media player supporting audio/video playback and basic library features. (Python)
- **Macropad** — Custom programmable keypad built with Arduino Pro Micro and QMK firmware for productivity macros. (Arduino, QMK)
- **Visual Novel** — Story-driven visual novel built with Ren'Py featuring branching dialogue and character art. (Ren'Py)
- **Snapchat Clone (Prototype)** — Unity prototype exploring ephemeral messaging and camera-first social features. (Unity)
- **Chatting App** — Real-time chat application deployed on Microsoft Azure with database-backed messaging. (Azure, Real-time)
- **E-commons Clothing** — Client WooCommerce website for an online clothing store with custom theming and product management. (WordPress, WooCommerce)
- **E-commerce (Games)** — WordPress + WooCommerce storefront built to sell indie games and digital products. (WordPress, WooCommerce)
- **Hero Academy** — React-based platform designed to help teachers manage tuition and student interactions efficiently. (React)

- **WakeelOnline** — React + TailwindCSS web platform connecting clients with verified lawyers and offering profile management features. (React, TailwindCSS)

Languages

- Urdu — Native
- English — Professional working proficiency

Interests

Competitive chess; rapid prototyping (hardware + software); manga and sci-fi reading

Links: Portfolio: <https://aka-shiro10.github.io/Sultan-Muhammad-Khan-Portfolio/> | GitHub: <https://github.com/aka-shiro10> | Email: akasultan10@gmail.com